Devemos ter em conta a ciência **quando ENSINAMOS CIÊNCIA?**

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TERESA E ALEXANDRE SOARES DOS SANTOS
INICIATIVA EDUCAÇÃO

Ideias muito difundidas em educação:	e muito modernas…
Ensino pela descoberta	J. Bruner (1961) – Dewey (1902) – Rousseau (1762)
Ensino por projetos	W.H. Kilpatrick (1918) – J. Dewey (1910)
Construção do conhecimento	Jean Piaget (1923) – Lev Vigotski (1926)

Memorização não: o importante é saber aprender

“As pessoas conhecedoras não são as que sabem tudo, mas as que sabem encontrar rapidamente a informação de que precisam”

“Educated people are not those who know everything, but rather those who know where to find, at a moment’s notice, the information they desire”

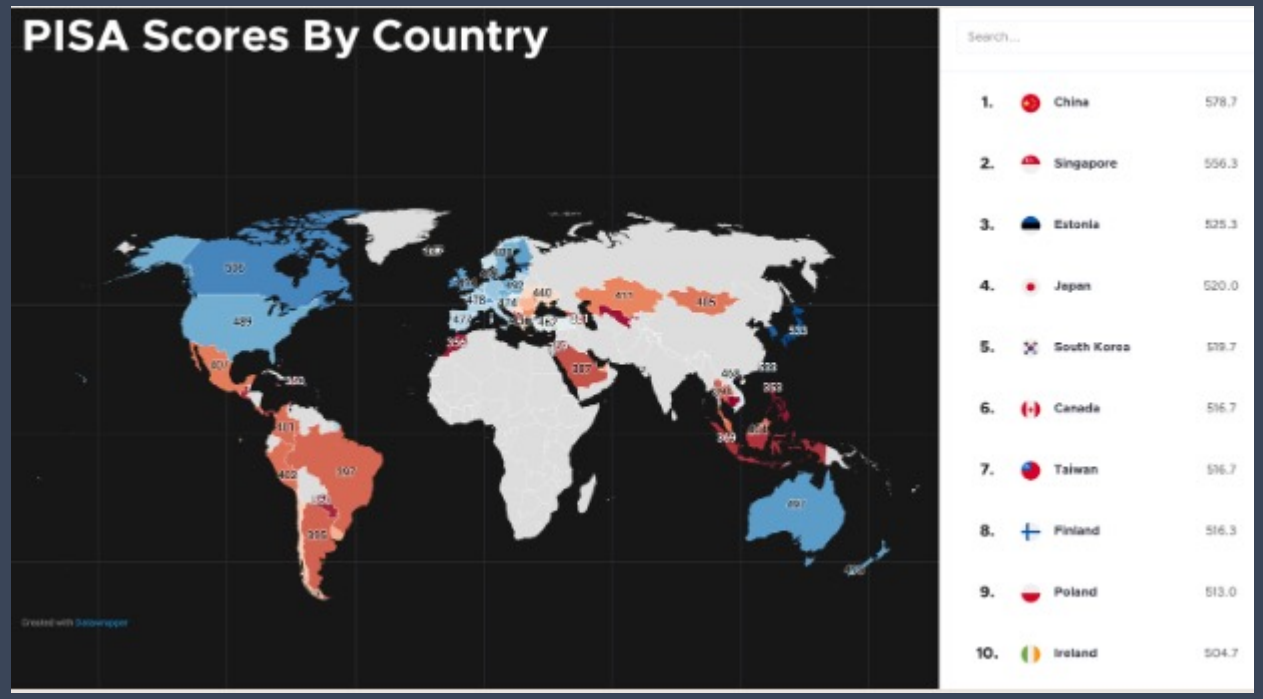
The Expositor and Current Anecdotes, Volume 16, *Indexing and Filing,* Page XX, Column 2, F. M. Barton, Publishing, Cleveland, Ohio, **1914-1915 !**

- Apelo à ignorância
- Falso: quanto mais se sabe mais se percebe
- Conhecimento enquadra conhecimento – consenso da psicologia cognitiva
- O conhecimento específico é importante – a transferência não é automática
- Ainda mais falso na atualidade

OCDE: o que diz e o que os dados mostram

“Education is no longer about teaching students something alone; it is more important to be teaching them to develop a reliable compass and the navigation tools”

“In the past, literacy was mainly about learning to read, a set of technical skills that individuals would acquire once for a lifetime in order to process an established body of coded knowledge. In the 21st century, literacy is about reading for learning…”



OECD (2019), *PISA 2018 Results (Volume I): What Students Know and Can Do*, PISA, OECD Publishing, Paris

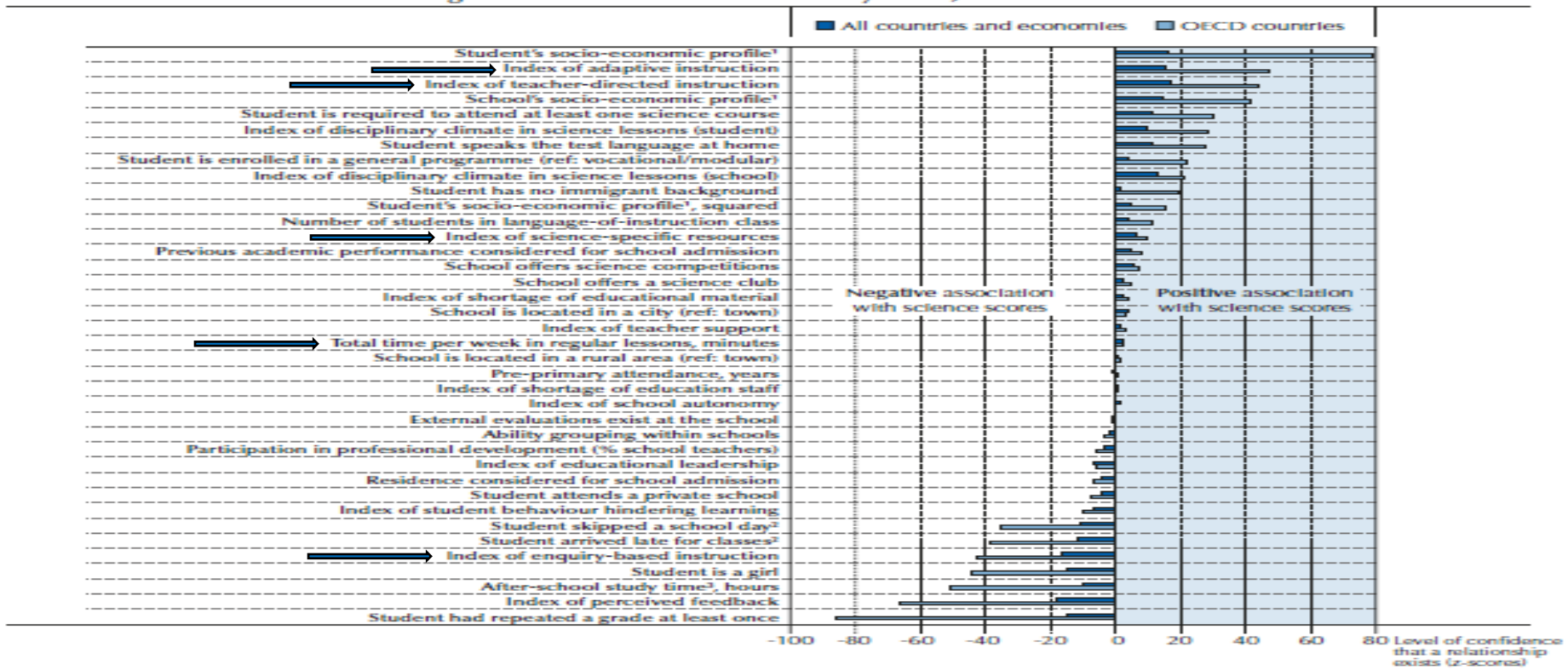
https://dx.doi.org/10.1787/9789264300002-en

Schleicher, A. (2012), Ed., *Preparing Teachers and Developing School Leaders for the 21st Century: Lessons from around the World*, OECD Publishing, Paris

Que revela o PISA?

Figure II.7.2 ▪ **Factors associated with science performance**

Multilevel regression models of education systems, schools and students



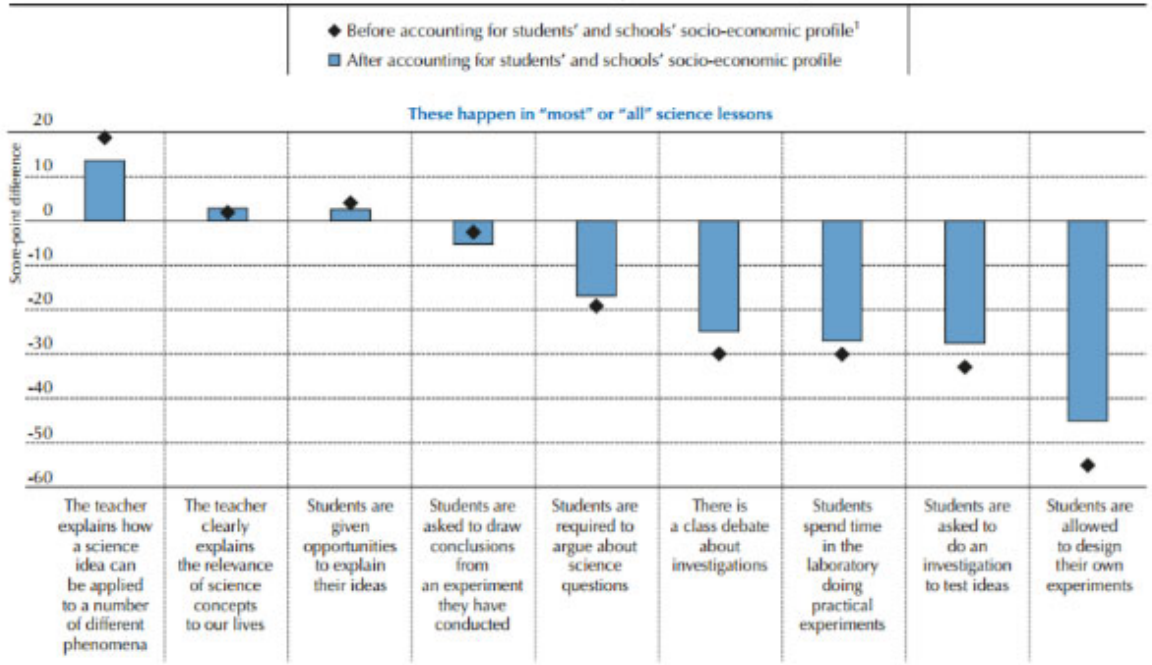
1. The socio-economic profile is measured by the PISA index of economic, social and cultural status (ESCS).
2. In the two weeks prior the PISA test.
3. Includes homework, additional instruction and private study.

Notes: All variables have been introduced jointly in a three-level regression model.
Statistically significant coefficients have associated z-scores below –1.96 or above 1.96.
The z-scores for «all countries and economies» are generally lower because the uncertainty surrounding the relationships is significantly higher.
See Table II.7.1. for results by education system.
Factors are ranked in descending order of the z-scores for OECD countries.
Source: OECD, PISA 2015 Database.

StatLink http://dx.doi.org/10.1787/888933436455

Figure II.2.20 ▪ **Enquiry-based teaching practices and science performance**

Results based on students' reports, OECD average



1. The socio-economic profile is measured by the PISA index of economic, social and cultural status.

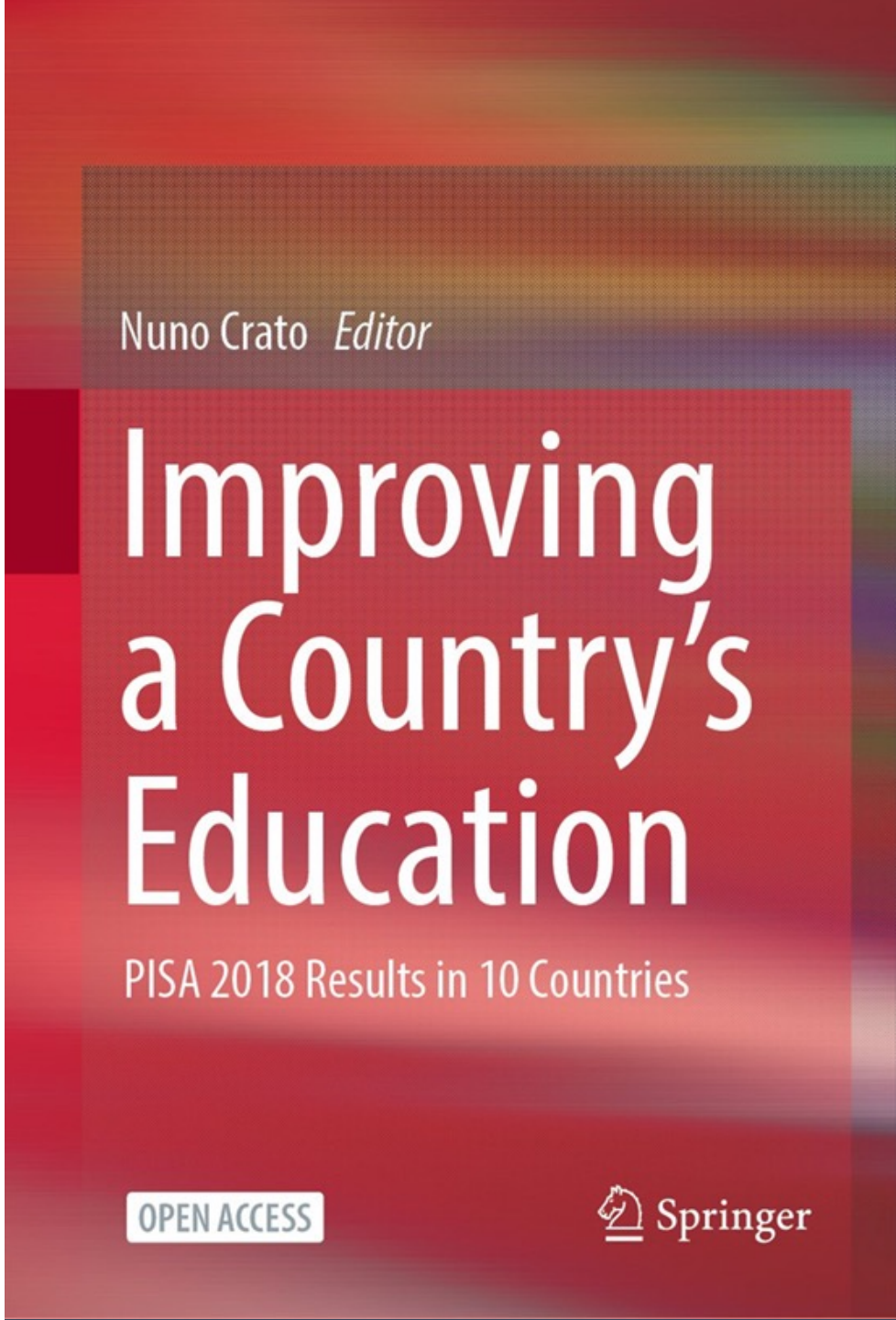
Note: All differences are statistically significant (see Annex A3).

Source: OECD, PISA 2015 Database, Table II.2.28.

StatLink http://dx.doi.org/10.1787/888933435628

Oliver, M., McConney, A. & Woods-McConney, A. The Efficacy of Inquiry-Based Instruction in Science: a Comparative Analysis of Six Countries Using PISA 2015. *Res Sci Educ* (2019). https://doi.org/10.1007/s11165-019-09901-0




Nuno Crato Editor
Improving a Country's Education
PISA 2018 Results in 10 Countries
OPEN ACCESS
Springer

Mas temos também informação da psicologia cognitiva moderna

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Estímulos

Os estímulos competem pela nossa atenção

Processa
≈ 4 informações em paralelo

Sistema sensorial

Memória de trabalho

Permite o processamento

Memória a longo prazo

Retém o que se pensa

Repite los esquemas

CLT: Teoria da carga cognitiva

Prestar atenção ao que se processa

Reference: 'Principles of Instruction Research-Based Strategies That All Teachers Should Know'
Barak Rosenshine, American Educator, 2012.



Barak Rosenshine

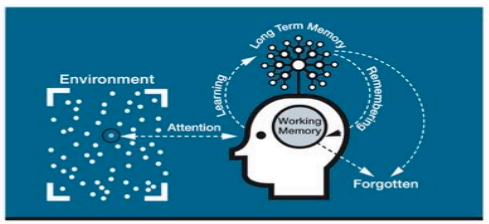
Principles of Instruction

Rosenshine's 10 Research-Based Strategies That All Teachers Should Know

Rosenshine's Principles of Instruction

10 Research-Based Classroom Strategies

Barak Rosenshine's 'Principles of Instruction' offers a comprehensive synthesis of decades of research, providing educators with ten foundational strategies to enhance teaching and learning. Grounded in cognitive science, observations of master teachers, and cognitive support research, these principles serve as a blueprint for effective instruction. The framework emphasises the significance of understanding cognitive limitations, applying incremental learning, and ensuring student mastery. It is crucial to understand that these strategies should not be rigidly used as a checklist but be applied in a nuanced manner.



'Simple Model of the Mind' interpreted by Oliver Caviglioli

The Limitations of Working Memory

Tailoring Instruction for Maximum Learning

At the heart of Rosenshine's Principles lies an understanding of the limitations of working memory, a crucial aspect of cognition where we process and temporarily store information. Our working memory has a limited capacity, capable of holding only a few pieces of information at any given moment. Overloading it with too much information can lead to confusion and hinder the learning process. This understanding underpins Rosenshine's advocacy for more explicit and instructional teaching and learning methods. For example, Rosenshine's research suggests should consider working memory limitations by presenting new material in manageable, small amounts, allowing students to master each concept before moving on to the next and asking questions to connect knowledge. These strategies ensure that students are not overwhelmed and can effectively process information meaningfully. Factors surrounding working memory are perfectly visualised in the above graphic by Oliver Caviglioli.



Daily Review

Review Previous Learning

Daily review aids students in connecting new concepts with existing knowledge, making learning more durable and meaningful. By engaging students in regular review, through activities like retrieval practice quizzes, homework correction tasks, and addressing common misconceptions as a class, students rehearse and enhance their skills to the point of automaticity. This form of daily practice makes it easier for students to acquire and assimilate future knowledge into their existing schemas.



Small Steps

Chunk New Information

Rosenshine explains that our working memory (the place where we process information) is small and therefore it can only process a few pieces of information at once. Because too much information swamps working memory, teachers should break down curriculum concepts and key processes into small manageable steps and allow for focused practice after each step. Breaking down information facilitates better understanding and gives teachers insights into students' knowledge gaps.



Questions

Check Students' Answers

Frequent questioning is an effective method of engaging students actively with new content and linking it to their existing knowledge. Utilising a broad range of questioning techniques (such as Doug Lemov's 'Cold Calling' or the 'Think, Pair, Share' routine) allows teachers to assess all students' understanding and promotes a culture of active participation. Questioning strategies also help to deepen students' grasp of the material by requiring them to apply and articulate their learning.



Models

Provide Cognitive Support

Rosenshine suggests that providing models and worked examples significantly aids students in problem-solving by offering essential cognitive support. This method, which includes teacher demonstrations and step-by-step explanations, helps reduce students' cognitive load. Effective modelling also includes think alouds where the teacher externalises their thinking process. Modeling is key in helping novices develop expertise by unveiling the expert's underlying thought processes.



Guided Practice

Support Student Practice

Guided practice is vital in the learning process as it ensures students actively engage with new material through sufficient rehearsal, crucial for transferring information to long-term memory. It involves the teacher spending significant time asking questions, checking understanding and correcting errors as students practice the new material. This iterative process of practice and feedback is essential for teachers in diagnosing any common misunderstandings and adjusting instruction accordingly.



Check Understanding

Address Misconceptions

Checking for understanding underscores the importance of regular assessment to ensure students grasp new material effectively. Frequent checks for understanding are critical, as they contribute to processing information into long-term memory and help teachers identify and rectify misconceptions early. Effective strategies include asking probing questions, encouraging students to summarise or explain concepts, and fostering discussions that require students to defend their specific viewpoints.



High Success Rate

Aim For 80% Success Rate

Ensuring students achieve a high success rate during instruction is crucial, with Rosenshine's research advocating for an optimal success rate around 80 percent. This benchmark balances challenge and attainability, indicating that students are not only learning the material successfully but are also being sufficiently challenged. Teachers can use tools such as mini-whiteboards to quickly check for levels of success from the entire class. If it is below 80%, it is recommended to reteach the material.



Scaffolds

Use Temporary Supports

Temporary supports assist students in mastering challenging concepts. These instructional scaffolds include strategies such as live modeling, think-alouds, sentence starters, and checklists, which are designed to guide students through the learning of new information. Effective teachers introduce these supports, gradually phasing them out as students gain proficiency. Scaffolds and supports ensure that students are neither overwhelmed by complexity nor under-challenged by simplicity.



Independent Practice

Build Fluency and Mastery

Following teacher-led instruction and guided practice, students should engage in independent practice to help reinforce their learning. Rosenshine explains that practice is essential for achieving fluency in skills and knowledge, allowing for automatic recall that frees up working memory for more complex cognitive tasks. Independent practice should involve the same material as the guided practice. During this process, teachers should actively monitor and circulate the classroom to maintain engagement.



Weekly/Monthly Review

Extend Practice Over Time

Rosenshine emphasises the importance of regular reviews, extending beyond daily checks to include weekly and monthly formative assessments. This advanced strategy, termed "successive relearning," involves spacing out retrieval practice over time to achieve mastery. Such reviews strengthen connections between new and old knowledge and ensures learning is transferred to long term memory. Effective retrieval practice tasks include cumulative quizzes, concept mapping, and class discussion.

Poster Designed by Jamie Clark | @XpatEducator | jamieleeclark.com

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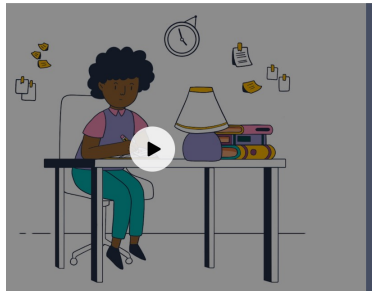


A Prática de Recuperação

Queres aprender melhor? Testa-te a ti próprio!

Depois de estudar tens a impressão de que sabes a matéria. Isto acontece porque a informação fica na tua memória recente. Quando revês, consegues reconhecer o que estudaste. Mas, atenção: reconhecer não é o mesmo que saber!

VER VÍDEO

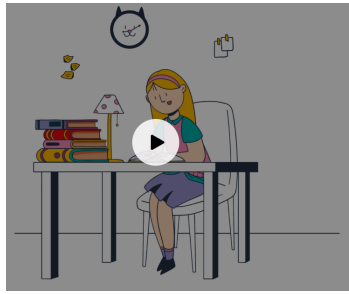


A Prática Espaçada

Tens tendência para esquecer a matéria que estudaste?

Isto acontece porque o estudo massivo, em curtos intervalos de tempo, não te ajuda a aprender. Quando estudas de véspera ou deixas tudo para a época dos exames, a aprendizagem vai ser sempre superficial. Organiza-te e faz pausas entre as matérias!

VER VÍDEO

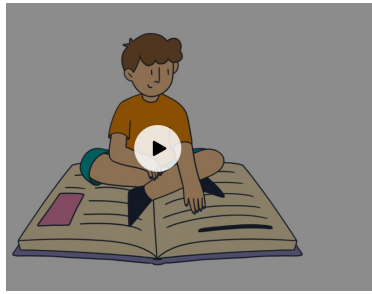


A Prática Intercalada

Em vez de insistires na mesma matéria, experimenta alternar entre temas diferentes. O segredo é intercalar!

Estudar o mesmo ponto da matéria durante muito tempo prejudica a tua capacidade de foco e facilita a divagação mental. Mas alternar conteúdos faz com que estejas sempre a receber novos estímulos: aumenta o desafio cognitivo e estimula a atenção e a persistência nas tarefas.

VER VÍDEO



A Prática de Elaboração

Sabias que aprendes mais se compreenderes melhor o que estudas?

Há uma estratégia de estudo que te pode ajudar a conseguir-lo. Chama-se prática da elaboração e exige que sejas capaz de explicar conceitos e descrever ideias de modo minucioso. Através da elaboração, associas o que estás a aprender ao que já sabes ou às experiências do dia-a-dia, melhorando a memória essa aprendizagem.

VER VÍDEO



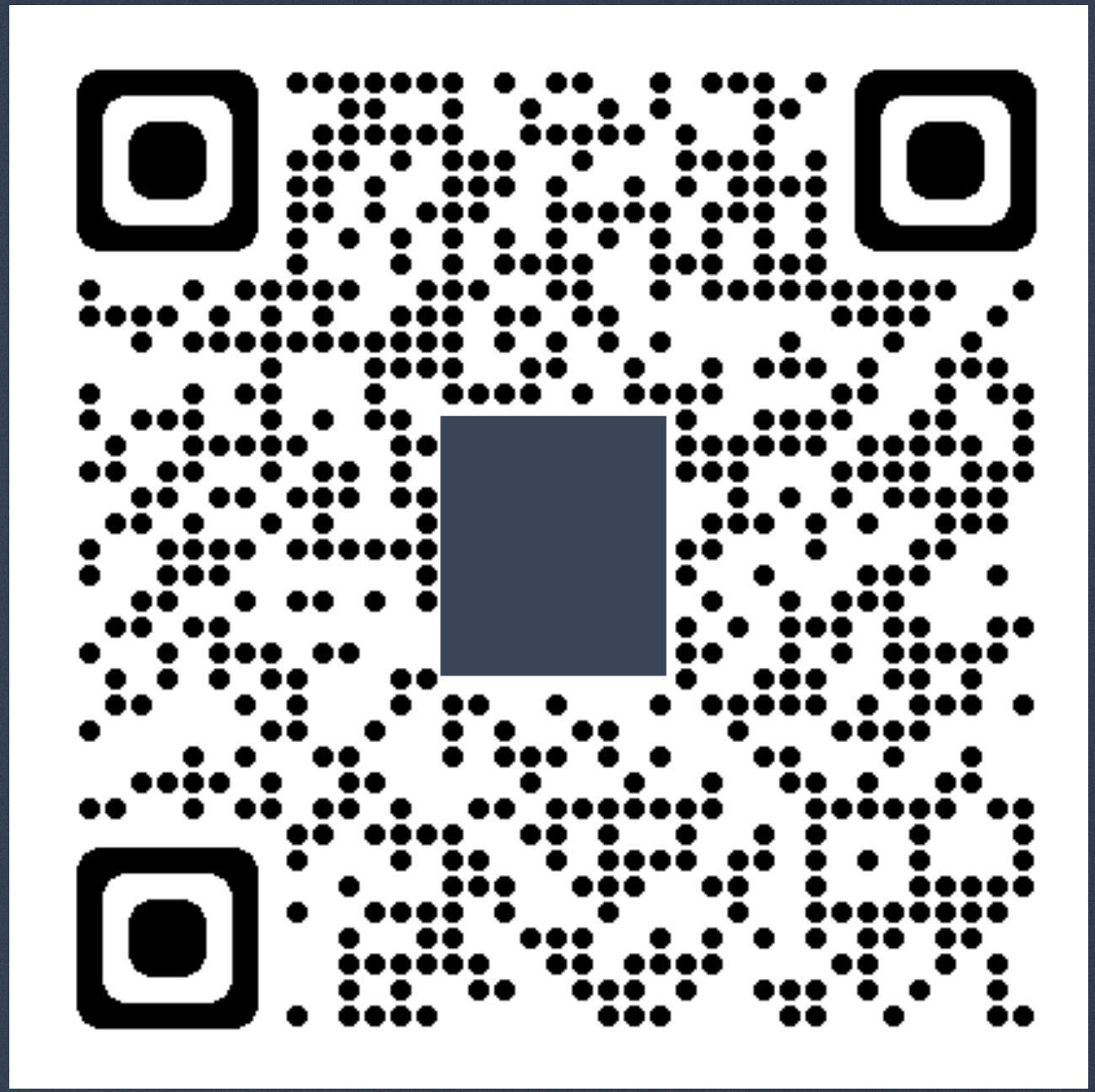
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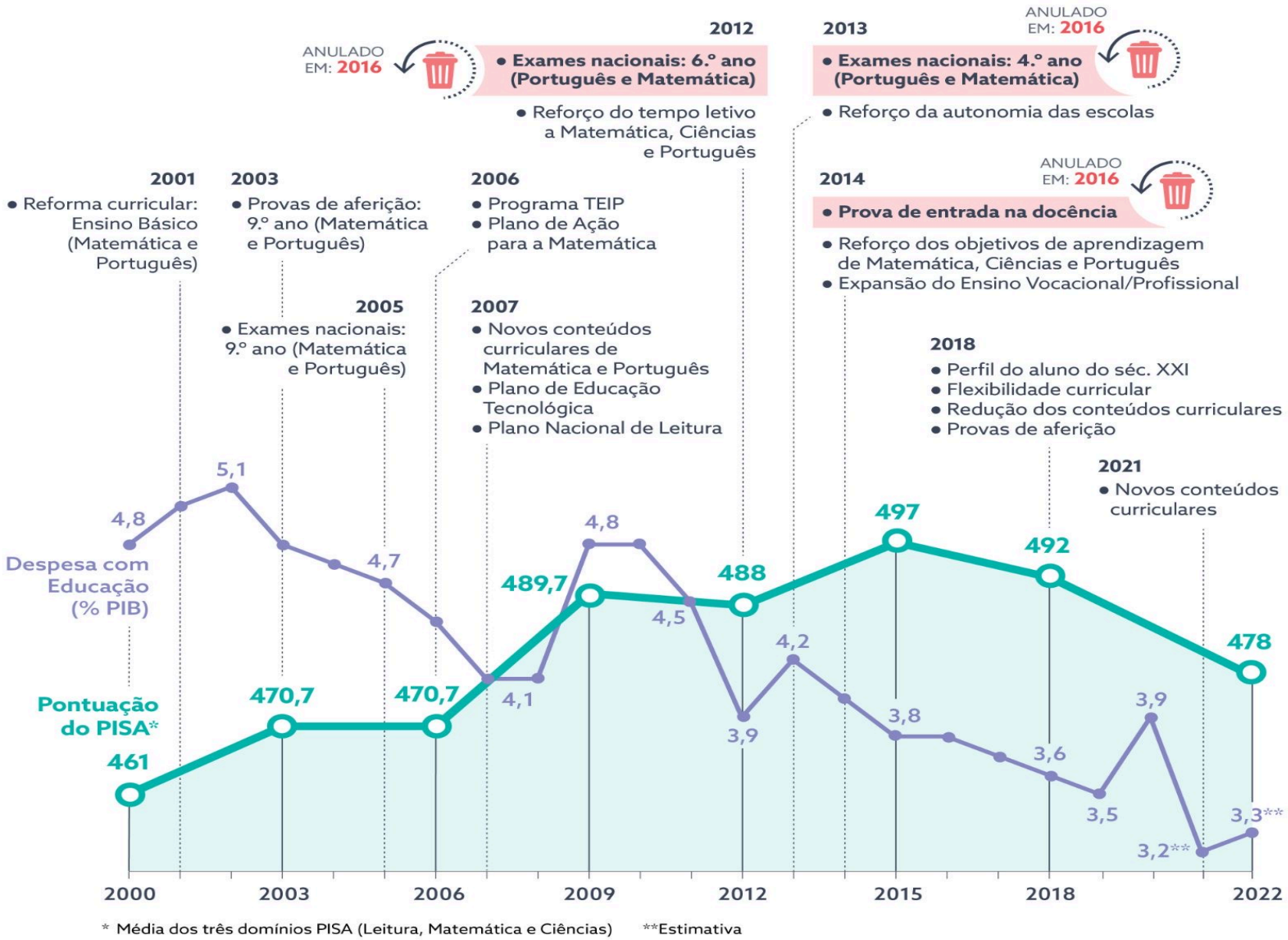


Obrigado!

Twitter
@eraty_num

POLÍTICAS EDUCATIVAS E RESULTADOS NO PISA

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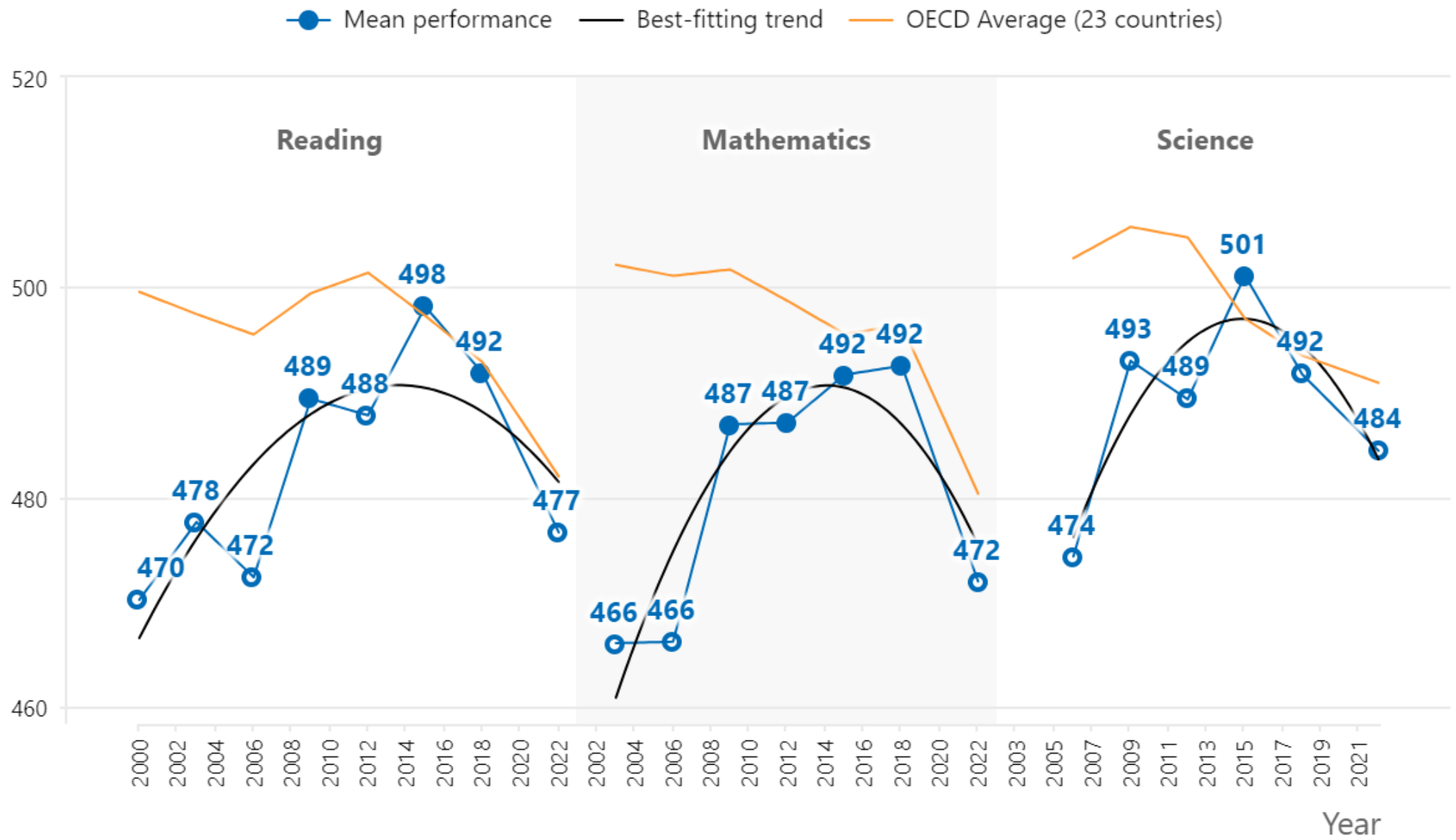
* Média dos três domínios PISA (Leitura, Matemática e Ciências) **Estimativa

Fonte: Marôco, 2021

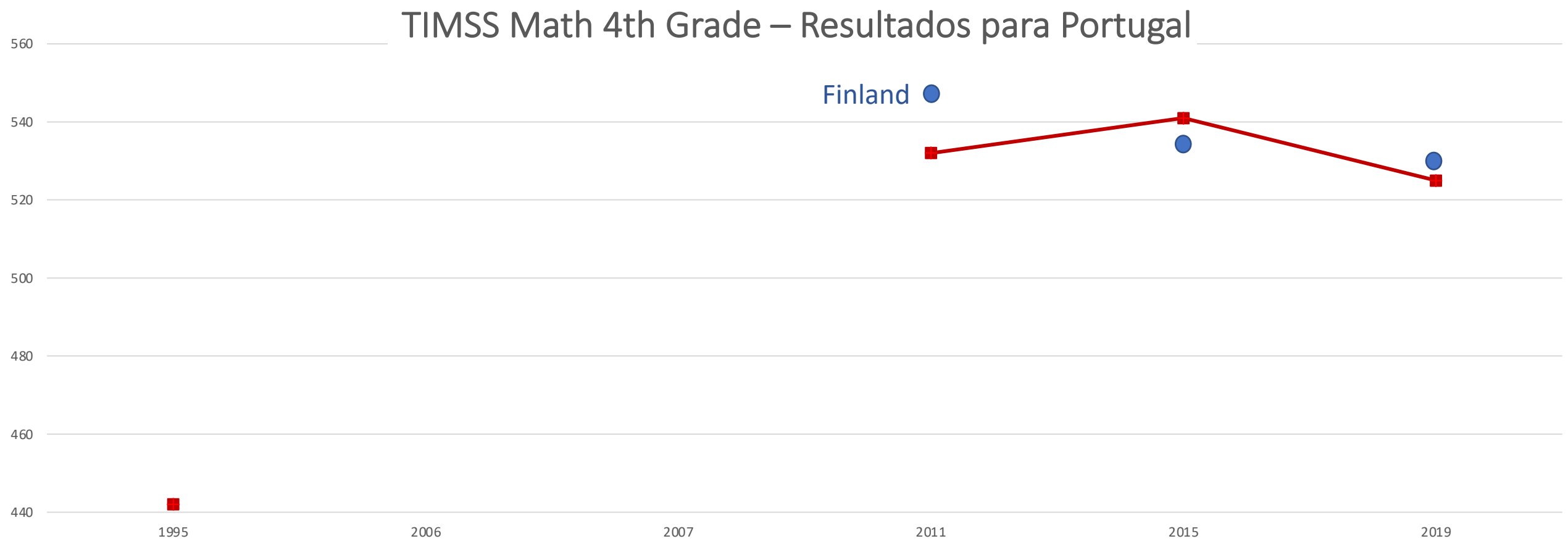
Infografia de Ana Serra

Portugal

Score points



era “romántica” 1974 - 1995/2000	tiempos pragmáticos 2000/2003 - 2011	conocimiento como base 2011 - 2015	competencias vagas… 2016 - 2023



Percentagem de high- vs low-performers

PISA - Portugal					
	2009	2011	2015	2018	2022
Science: High-performers	4.2	4.5	7.4	5.6	5.0
Low-performers	16.5	19.0	17.4	20.2	22.0
Maths: High-performers	9.6	10.6	11.4	11.6	7.0
Low-performers	23.7	24.9	23.8	23.3	30.0
Reading: High-performers	4.8	5.8	7.5	7.3	5.0
Low-performers	17.6	18.8	17.2	19.6	23.0

High performers > 4; Low Performers < Level 2

TIMSS 4th Grade Math - Portugal			
	2011	2015	2019
High Performers	8	12	9
Low Performers	20	18	26

High performers = level 4; Low Performers ≤ Level 1

2003-2015

Mais ambição curricular

Objetivos mais precisos

Mais avaliação

2016-2022

Menor ambição curricular

Objetivos desvalorizados

Maior flexibilidade curricular

Menos avaliação